

Type: MCQ

Q1. What is the maximum number of divisions made by Euclid's algorithm for the inputs m & n for any values in the range $1 < m < n \leq 10$? (0.5)

1. ** 5
2. 2
3. 6
4. 3

Q2. Identify the pair of functions in which the first function's order of growth is higher than the second function: (0.5)

1. $n(n + 1)$ and $2000n^2$
2. $(n - 1)!$ and $n!$
3. 2^{n-1} and 2^n
4. ****** $\log^2_2(n)$ and $\log_2(n^2)$

Q3. What will be the output list after completing the first pass of bubble sort on the input array 32, 51, 27, 85, 66, 23, 13, 57 ? (0.5)

1. 23, 13, 27, 33, 51, 57, 66, 85
2. 32, 51, 27, 66, 23, 13, 57, 85
3. 27, 33, 51, 23, 13, 57, 66, 85
4. ** 32, 27, 51, 66, 23, 13, 57, 85

Q4. Which of the following can be stated as the problem of finding the shortest Hamiltonian Circuit of the graph?
(0.5)

1. Assignment problem
2. ******Traveling Salesman problem
3. Breadth first search
4. Depth first search

Q5. How many character comparisons are made by the brute force algorithm in searching for the pattern **RRRRS** in the text

RRRRRRRRRRRRRRRRRRRRRRRRRS

(0.5)

1. ** 105
2. 100
3. 21
4. 25

Q6. In what order should we insert the elements $\{1, 2, 3, 4, 5, 6, 7\}$ into an empty AVL tree so that we don't have to perform any rotations on it?

(0.5)

1. 4,2,1,6,3,5,7
2. 4,2,6,1,3,5,7
3. **4,2,1,6,3,7,5
4. 4,1,2,6,3,7,5

Q7. Which of the following stable sorting algorithm takes the least time when applied to an almost sorted array?
(0.5)

1. Quick sort
2. Insertion sort

3. Selection sort
4. ****** Merge sort

Q8. Consider the problem of searching an element (x) in an array arr[] of size (n). The problem can be solved in $O(\log n)$ time if

- I. Array is sorted
- II. Array is sorted and rotated by (k), where (k) is given and $k \leq n$
- III. Array is not sorted

(0.5)

1. I only
2. ******I and II only
3. I, II and III only
4. None of the mentioned

Q9. In the context of an extended Binary tree (0.5)

- I. The extension of the empty binary tree is a single external node.
 - II. The number of internal nodes is always 1 more than the number of external nodes.
1. Both are True
 2. Both are False
 3. ******I – True, II – False
 4. I – False, II – True

Q10. When a graph is traversed using BFS method, if a new unvisited vertex v is reached for the first time from a current vertex say u, then the edge (u,v) is called as _____ (0.5)

1. ****** Tree Edge
2. Cross Edge
3. Back Edge
4. Solid Edge

11)

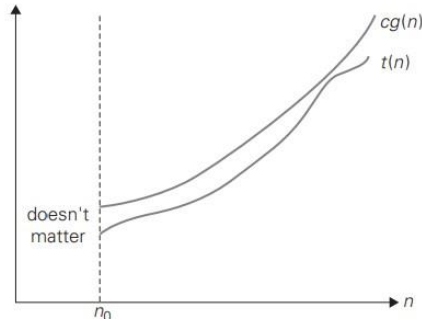
There are mainly three asymptotic notations:

1. Big-O notation
2. Omega notation
3. Theta notation

Big-O notation: A function $t(n)$ is said to be in $O(g(n))$, denoted $t(n) \in O(g(n))$, if $t(n)$ is bounded above by some constant multiple of $g(n)$ for all large n , i.e., if there exist some positive constant c and some nonnegative integer n_0 such that

0.5 Marks

$$t(n) \leq cg(n) \text{ for all } n \geq n_0.$$



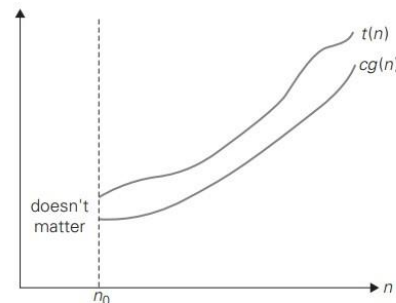
Example: $100n + 5 \leq 100n + n$ (for all $n \geq 5$) $= 101n \leq 101n^2$

0.5 Marks

Ω (Omega) notation : A function $t(n)$ is said to be in $\Omega(g(n))$, denoted $t(n) \in \Omega(g(n))$, if $t(n)$ is bounded below by some positive constant multiple of $g(n)$ for all large n , i.e., if there exist some positive constant c and some nonnegative integer n_0 such that

0.5 Marks

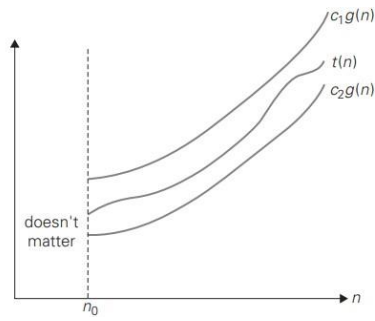
$$t(n) \geq cg(n) \text{ for all } n \geq n_0.$$



Example : $n^3 \in (n^2): n^3 \geq n^2$ for all $n \geq 0$

0.5 Marks

Θ (Theta) notation : A function $t(n)$ is said to be in $\Theta(g(n))$, denoted $t(n) \in \Theta(g(n))$, if $t(n)$ is bounded both above and below by some positive constant multiples of $g(n)$ for all large n , i.e., if there exist some positive constants c_1 and c_2 and some nonnegative integer n_0 such that



$c_2g(n) \leq t(n) \leq c_1g(n)$ for all $n \geq n_0$.

0.5 Marks

Example :

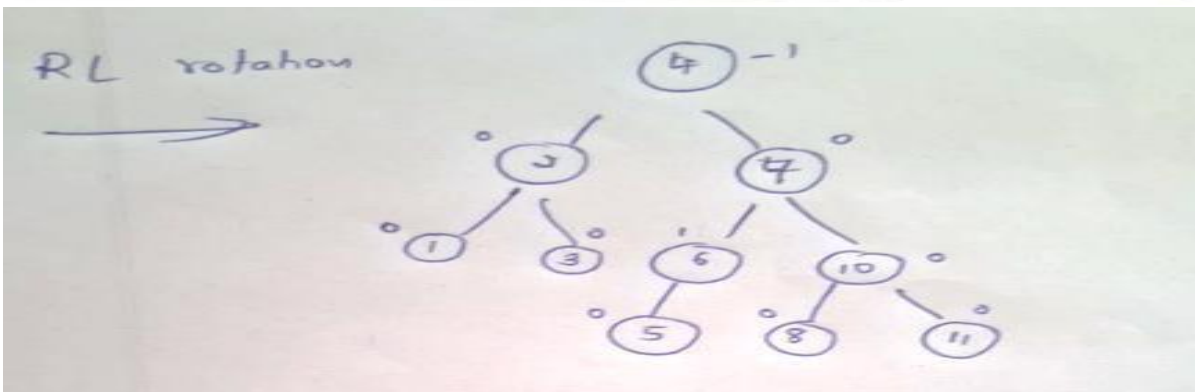
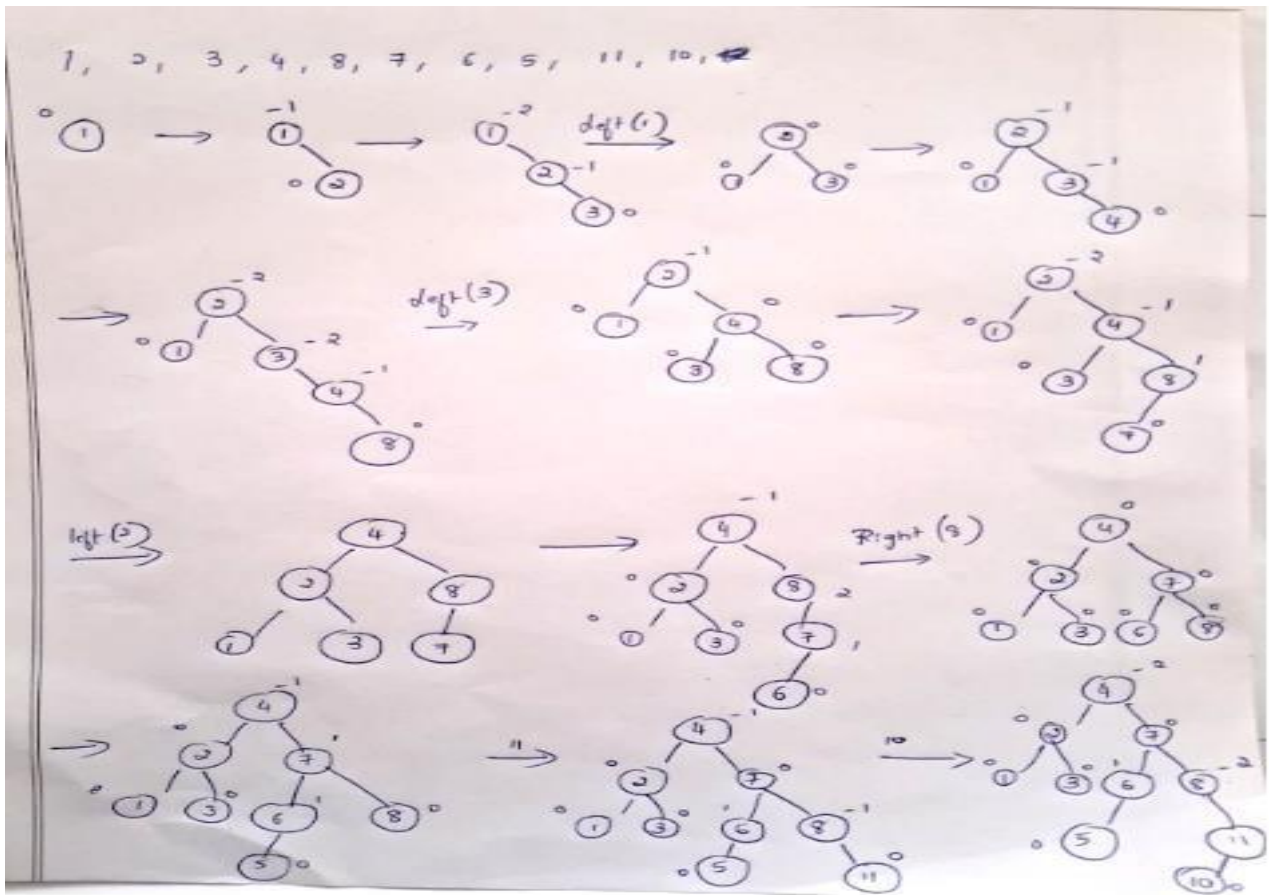
0.5 Marks

Note: marks meaning to written the

$$\frac{1}{2}n(n-1) = \frac{1}{2}n^2 - \frac{1}{2}n \leq \frac{1}{2}n^2 \quad \text{for all } n \geq 0.$$

have giving which has closer the definition and one who not map.

12)



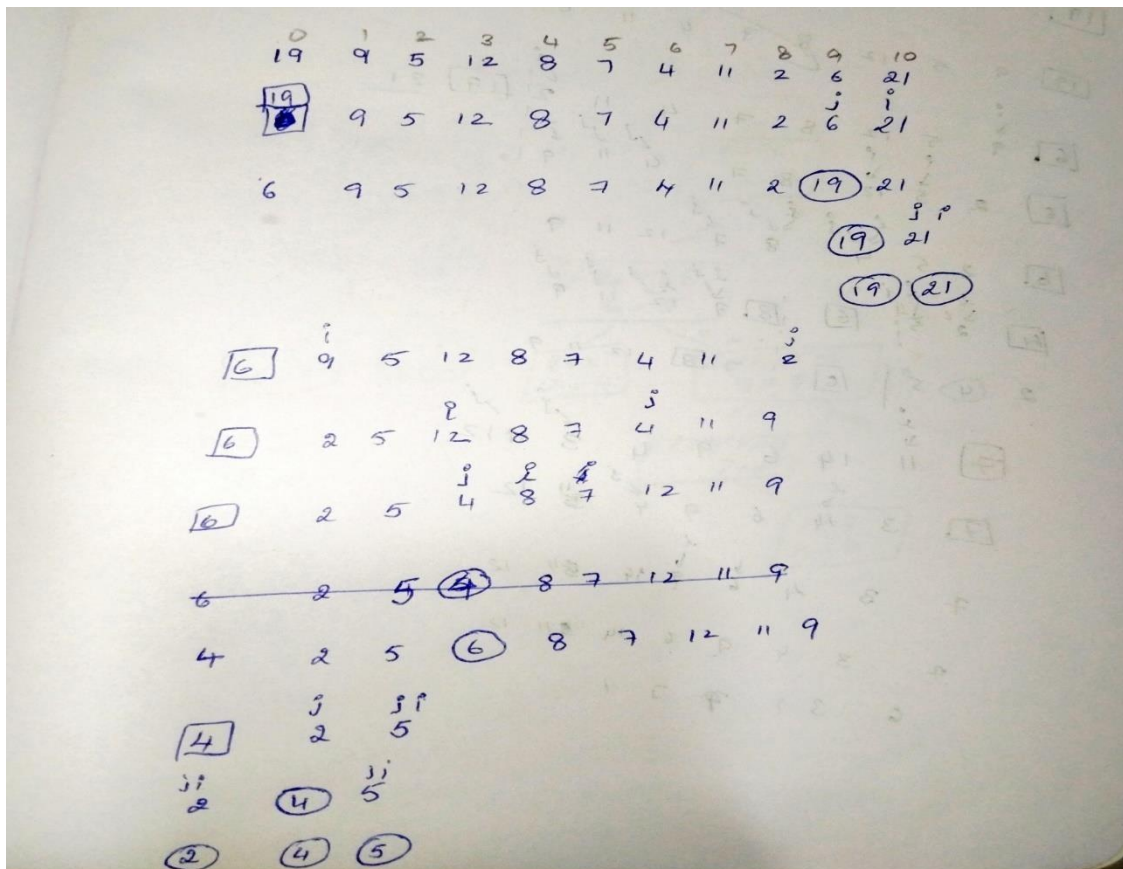
13) Apply quicksort to sort the list: 19, 9, 5, 12, 8, 7, 4, 11, 2, 6, 21 in ascending order. Draw the tree of the recursive calls made.

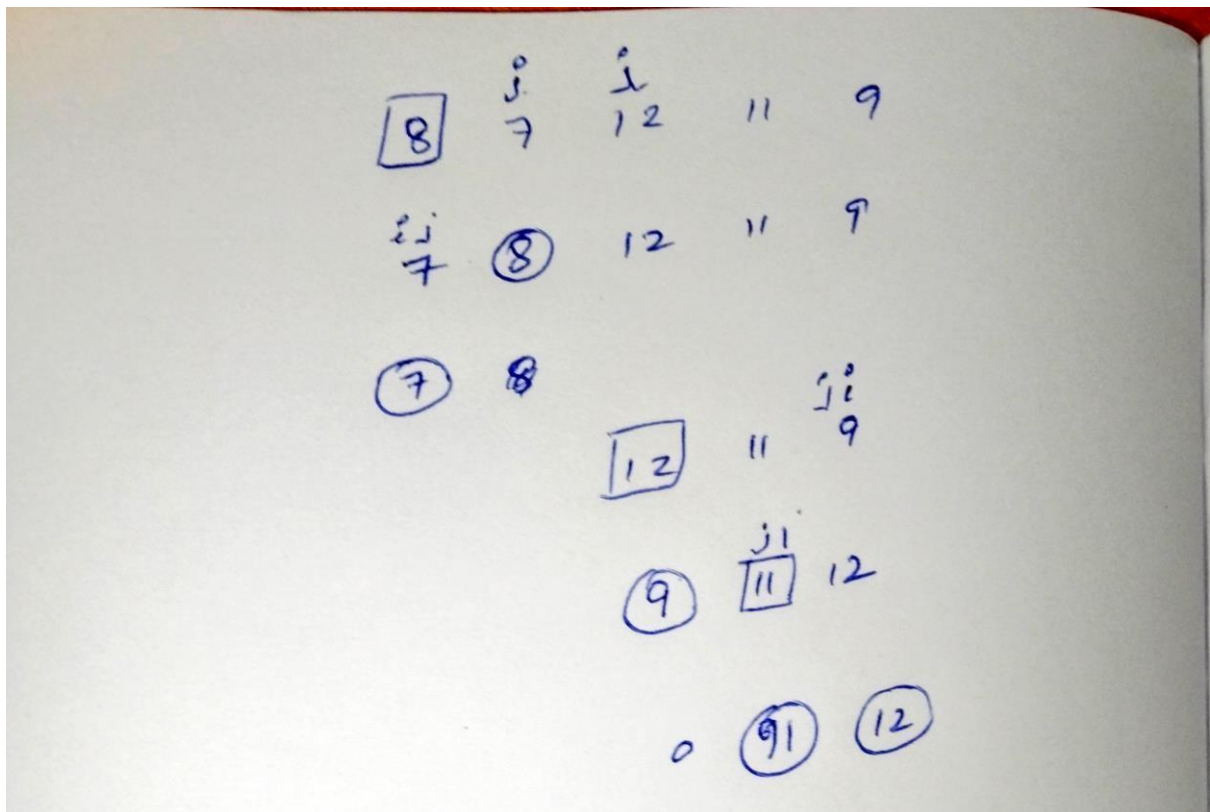
If any other algorithm other than Hoare is used for partition, then the pseudocode of the algorithm must be written.

All the steps need to be written clearly.

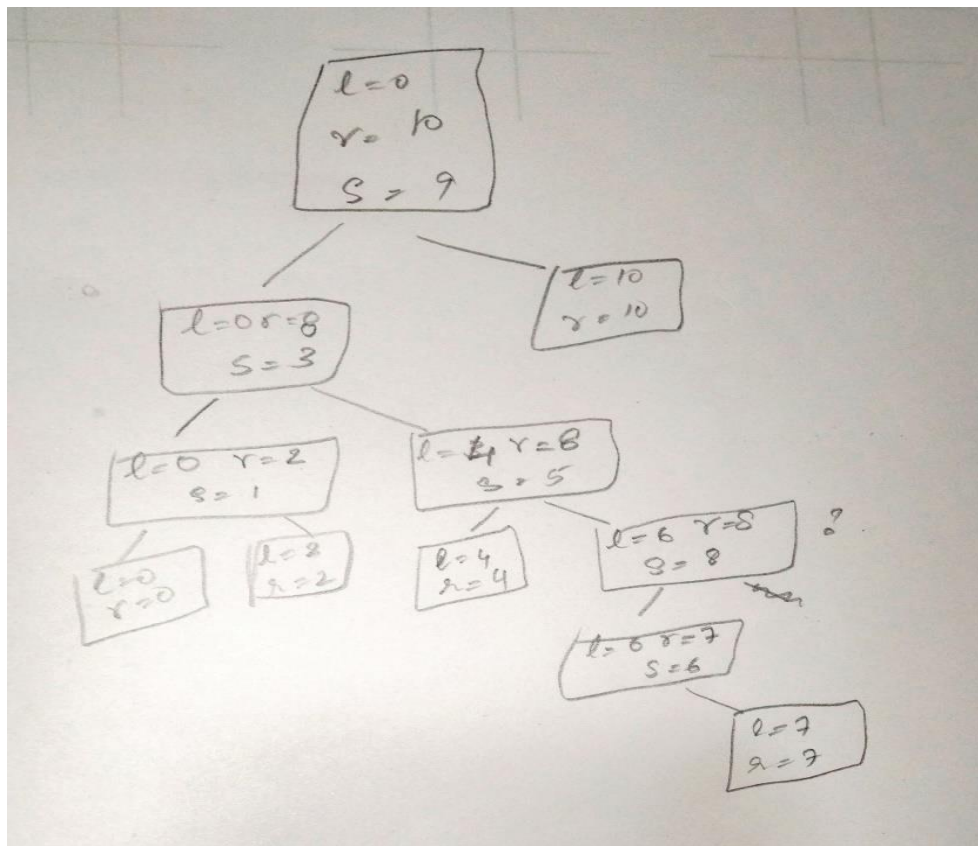
Refer to the textbook for the format of the tree of the recursive calls.

✦ All the split points needs to be clearly written





1.5M



1M

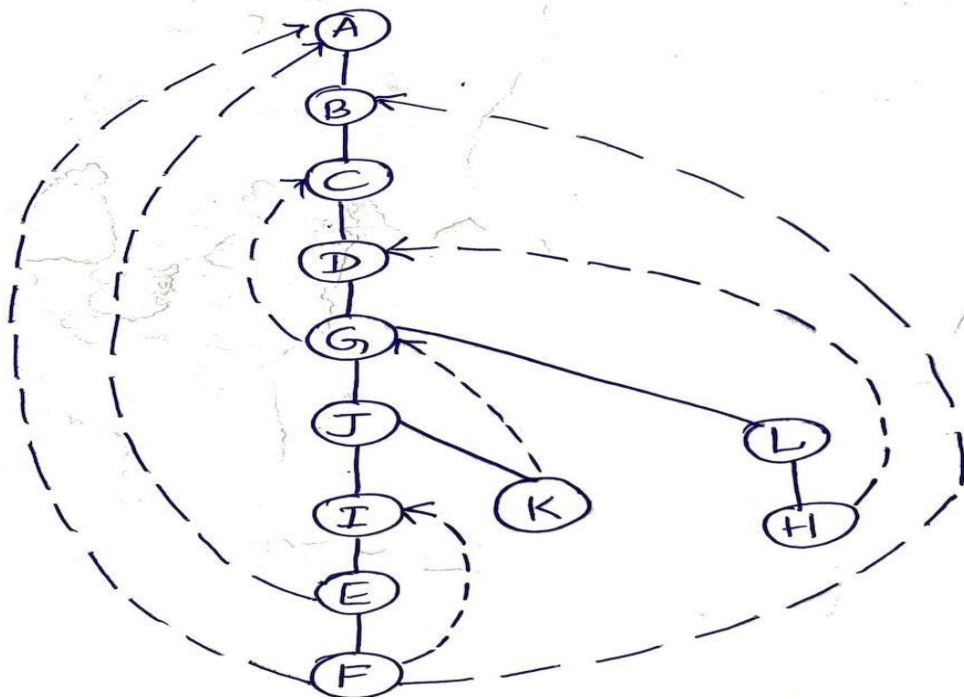
Q14) Push order of vertices - 1M

A B C D G J I E F K L H

Pop order of vertices - 1M

F E I K J H L G D C B A

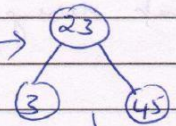
DFS TREE - 1M

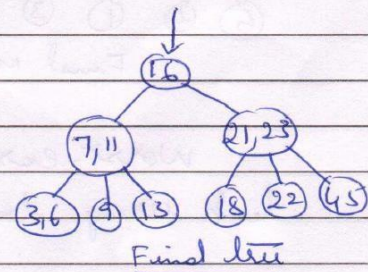
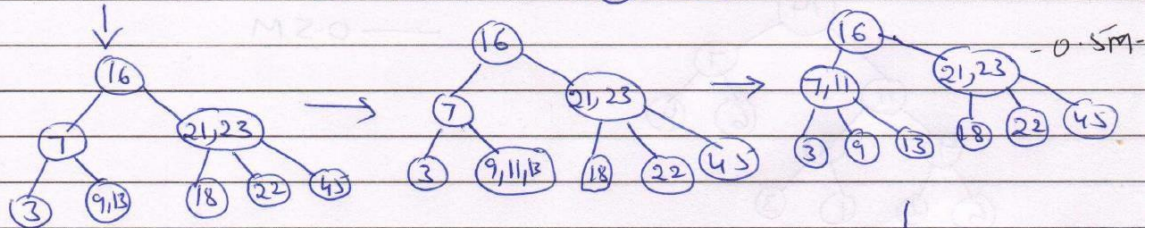
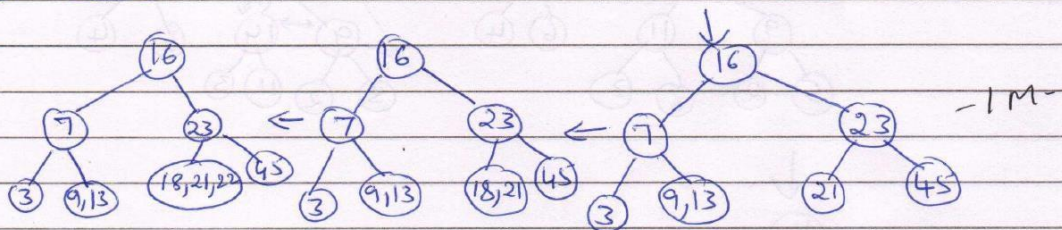
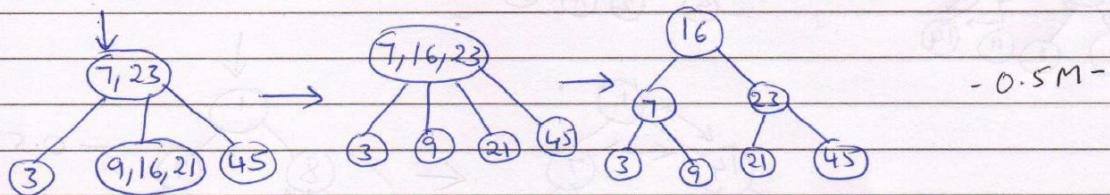
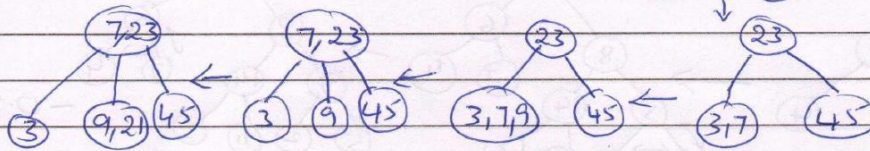


If alphabetical order is not followed marks have been deducted. Whichever part of answer is missing marks are deducted accordingly.

15) 2-3 Tree

23, 45, 3, 7, 9, 21, 16, 13, 18, 22, 11, 6

(23) → (23, 45) → (3, 23, 45) →  - 0.5M -



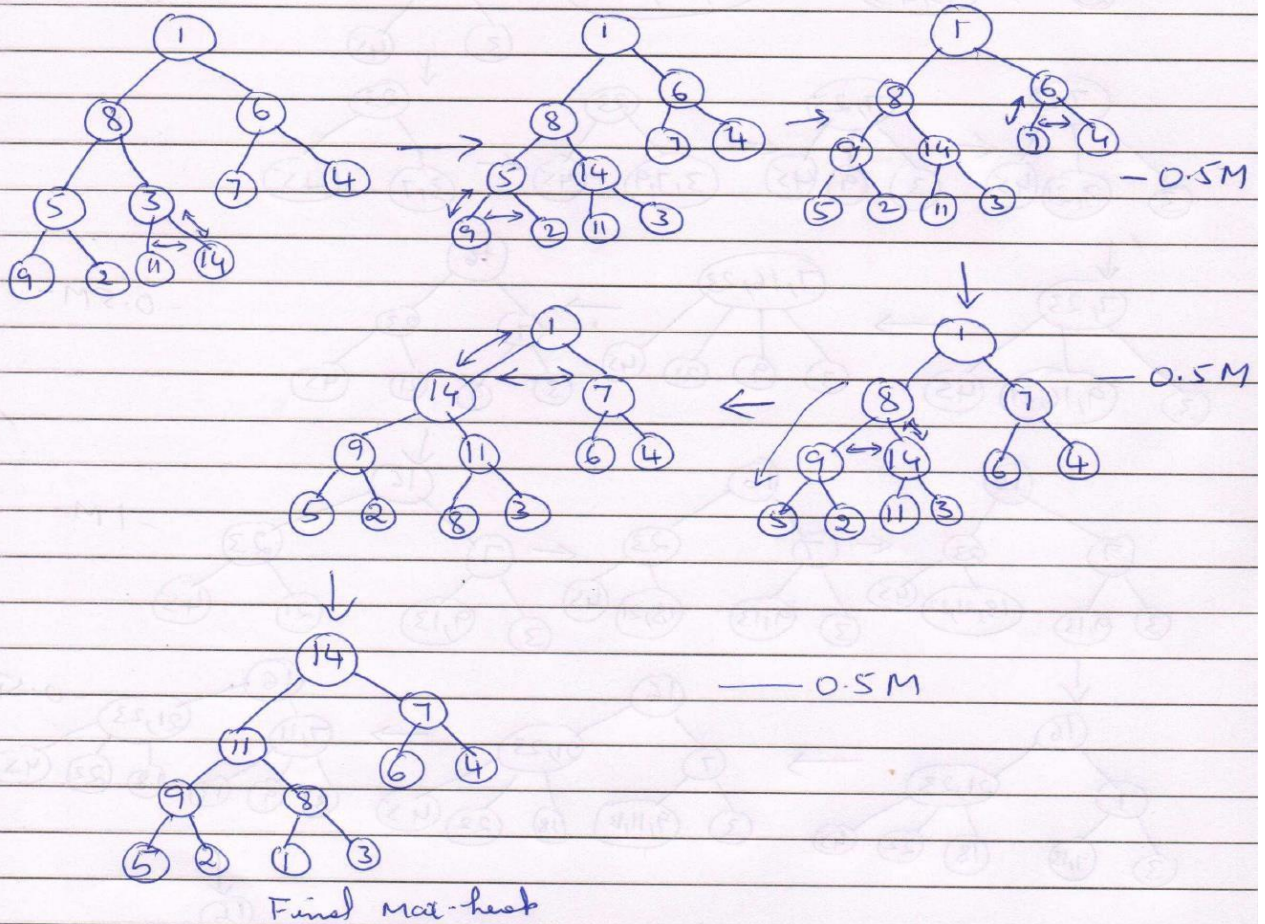
The maximum number of comparisons in the full tree of height 3 is = 8 [Every level 2 comparisons $\times 4 = 8$]

- 0.5M -

16

1, 8, 6, 5, 3, 7, 4, 9, 2, 11, 14

Max-heap using bottom-up algorithm



Worst-case time complexity of heap construction using heap using bottom up algorithm:

$$C_{\text{worst}}(n) = \sum_{i=0}^{h-1} \leq 2(h-i) = \sum_{i=0}^{h-1} 2(h-i)2^i \quad -0.5M$$

$$= 2 \left[\sum_{i=0}^{h-1} h2^i - \sum_{i=0}^{h-1} i \cdot 2^i \right]$$

$$= 2 \left[h \sum_{i=0}^{h-1} 2^i - \sum_{i=0}^{h-1} i \cdot 2^i \right]$$

$$= 2 \left[h(2^h - 1) - [(h-2)2^h + 2] \right]$$

$$= 2 \left[h(2^h - 1) - h2^h + 2 \cdot 2^h - 2 \right]$$

$$= 2 \left[\cancel{h2^h} - h - \cancel{h2^h} + 2 \cdot 2^h - 2 \right]$$

$$= 2 \left[-h + 2 \cdot 2^h - 2 \right]$$

$$= 2 \left[-h + 2^{h+1} - 2 \right]$$

$$= 2 \left[-h - 1 + \underbrace{2^{h+1} - 1}_{\rightarrow n} \right]$$

$$[\because n = 2^{h+1} - 1]$$

$$= 2 \left[-\log_2(n+1) + n \right]$$

$$= 2 \left[n - \log_2(n+1) \right]$$

$$C_{\text{major}}^{(n)} \in O(n)$$

0.5M

0.5M

Q17 Write the general procedure of Boyer-Moore string matching algorithm. Trace the same to search for a pattern **RIMJIM** in the text: **THE-MUSIC-SOUNDS-RIMJIMSE**

Also find the number of character comparisons. (3m)

The Boyer-Moore algorithm

- Step 1** For a given pattern and the alphabet used in both the pattern and the text, construct the bad-symbol shift table as described earlier.
- Step 2** Using the pattern, construct the good-suffix shift table as described.
- Step 3** Align the pattern against the beginning of the text.
- Step 4** Repeat the following step until either a matching substring is found or the pattern reaches beyond the last character of the text. Starting with the last character in the pattern, compare the corresponding characters in the pattern and the text until either all m character pairs are matched (then stop) or a mismatching pair is encountered after $k \geq 0$ character pairs are matched successfully. In the latter case, retrieve the entry $t_1(c)$ from the c 's column of the bad-symbol table where c is the text's mismatched character. If $k > 0$, also retrieve the corresponding d_2 entry from the good-suffix table. Shift the pattern to the right by the number of positions computed by the formula

$$d = \begin{cases} d_1 & \text{if } k = 0 \\ \max\{d_1, d_2\} & \text{if } k > 0 \end{cases}, \quad (7.3)$$

where $d_1 = \max\{t_1(c) - k, 1\}$.

1 mark

17) Bad symbol shift table.

R	I	M	J	others
5	1	3	2	6

{0.5 M}

Good Suffix shift table

k		
1	RIMJIM	6
2	RIMJIM	3
3	RIMJIM	6
4	RIMJIM	6
5	RIMJIM	6

{0.5 M}

THE-MUSIC-SOUNDS-RIMJIMSE

RIMJIM
($k=0, d_1=6$)

RIMJIM
($k=0, d_1=6$)

RIMJIM

($k=0, d_1=5$)

RIMJIM

[0.5 M]

No. of comparisons = $1+1+1+6 = \underline{\underline{09}}$

[0.5 M]

Q18)

Input: b, c, d, c, b, a, a, b, c, a, b

Frequency

a	b	c	d
3	4	3	1

Distribution values

a	b	c	d
3	7	10	11

0.5

Ans

A[10] =

	a	b	c	d
b	3	7	10	11
a	3	6	10	11
c	2	6	10	11
b	2	6	9	11
a	2	5	9	11
a	1	5	9	11
b	0	5	9	11
c	0	4	9	11
d	0	4	8	11
c	0	4	8	10
b	0	4	7	10

S[0..10]	0	1	2	3	4	5	6	7	8	9	10
							b				
			a								
										c	
						b					
		a									
	a										
						b					
										c	
											d
									c		
					b						
	a	a	a	b	b	b	b	c	c	c	d

1.5 M